

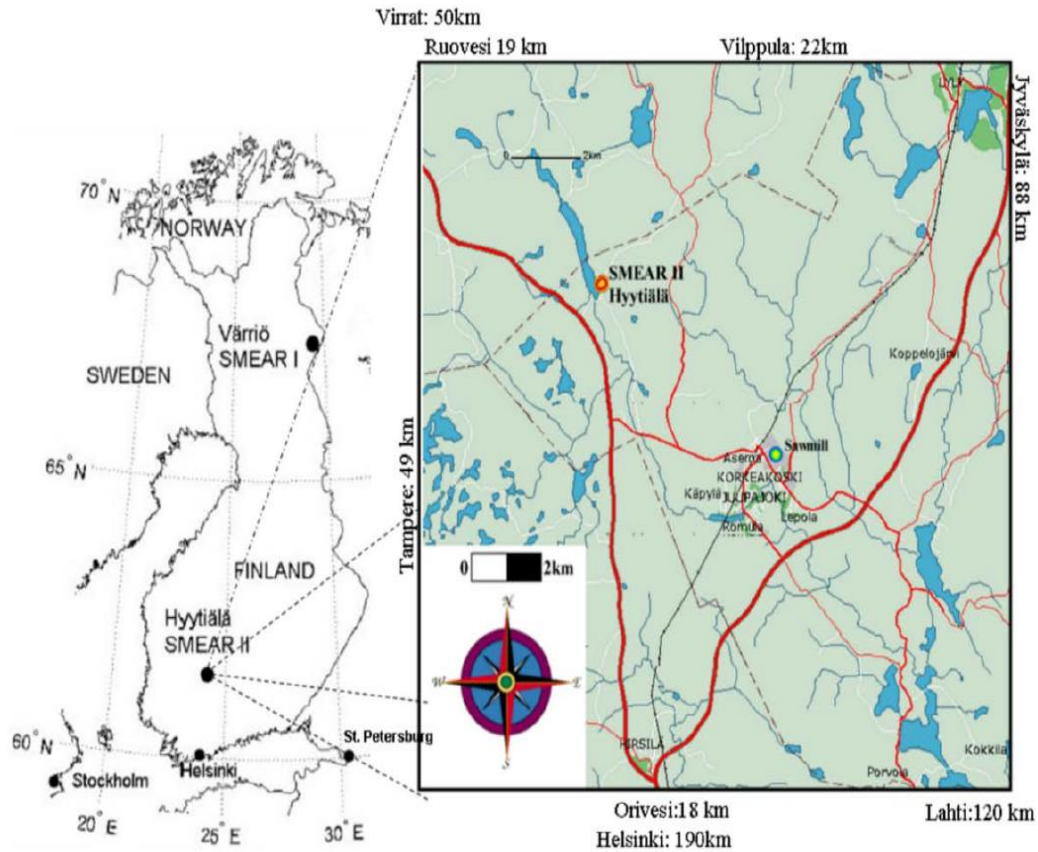
# **Effect of dust and pollen events on optical properties, size distributions, and mass concentrations of aerosol particles in a Finnish boreal forest**

Sujai Banerji<sup>1</sup>, Touqeer Gill<sup>1</sup>, Veli-Matti Kerminen<sup>1</sup>, and Tuukka Petäjä<sup>1</sup>

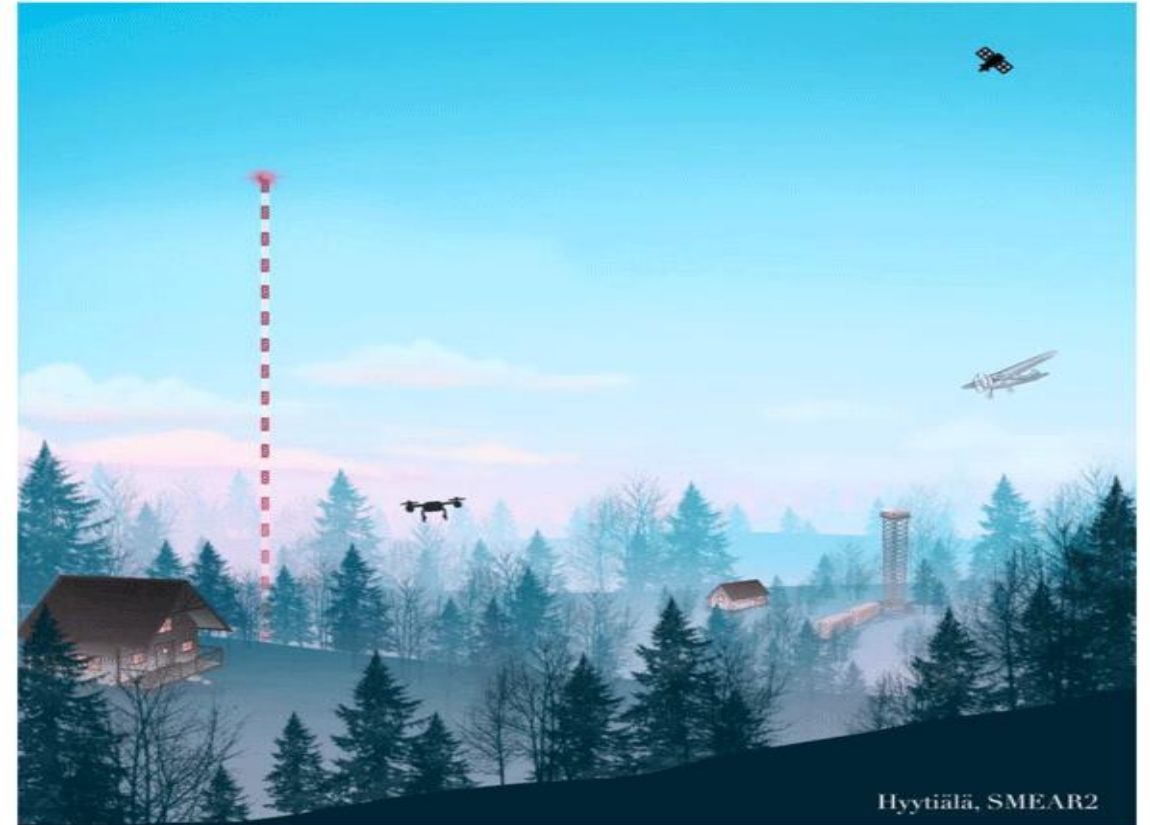
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# Study region



Eerdekens et al., 2009



Brasseur et al., 2022

# Instruments and measurement setup

Multi-angle absorption  
photometer (Thermo  
Scientific MAAP  
Model 5012)

Aethalometer (Magee  
Scientific model AE33)

PM<sub>10</sub>/PM<sub>1</sub> switching  
inlet

PM<sub>10</sub> range = [(10, 19), (30, 39), (50, 59)]

PM<sub>1</sub> range = [(0, 9), (20, 29), (40, 49)]

Integrating  
nephelometer  
TSI model 3563

Introduction

Methodology

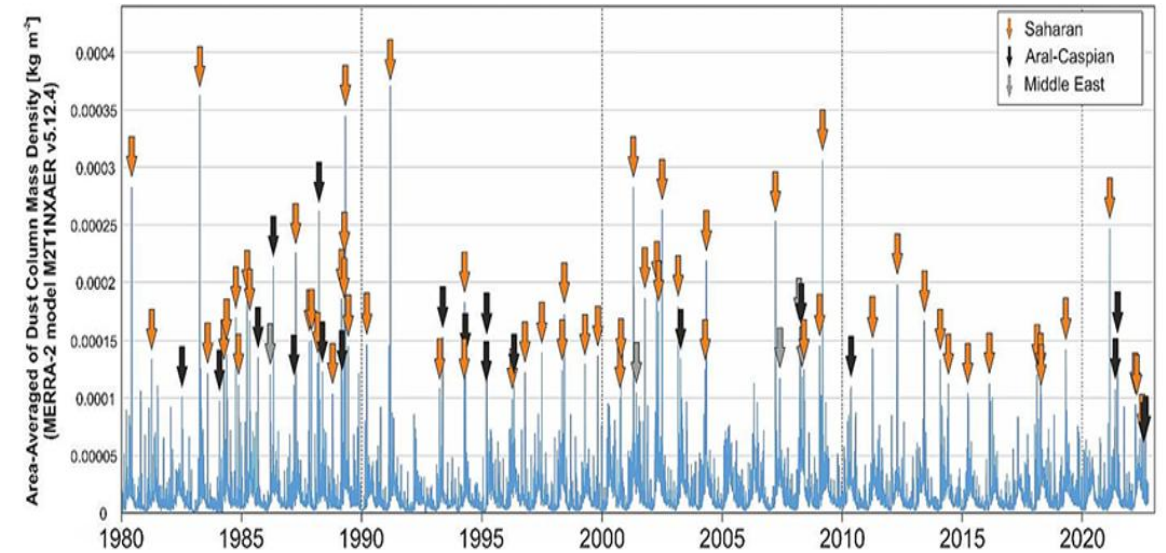
Results and discussion

Conclusion

# Long-range transport of dust to Finland (1980-2022)

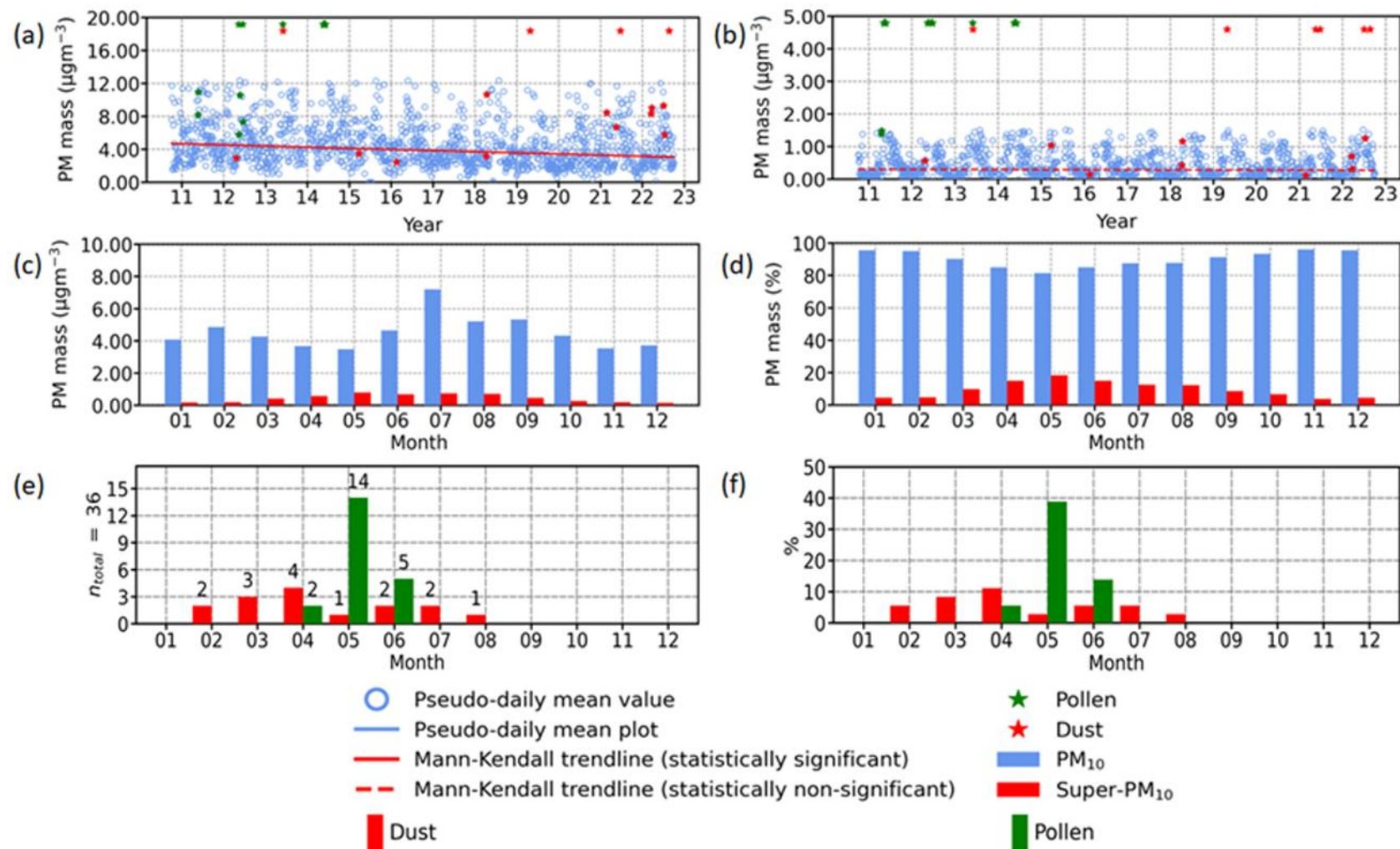


Varga et al, 2023



Varga et al, 2023

# Pollen events in Finland



Banerji et al, 2025

# Research objectives

1

Effect of dust, and pollen events on optical properties, size distributions, and mass concentrations of aerosol particles.

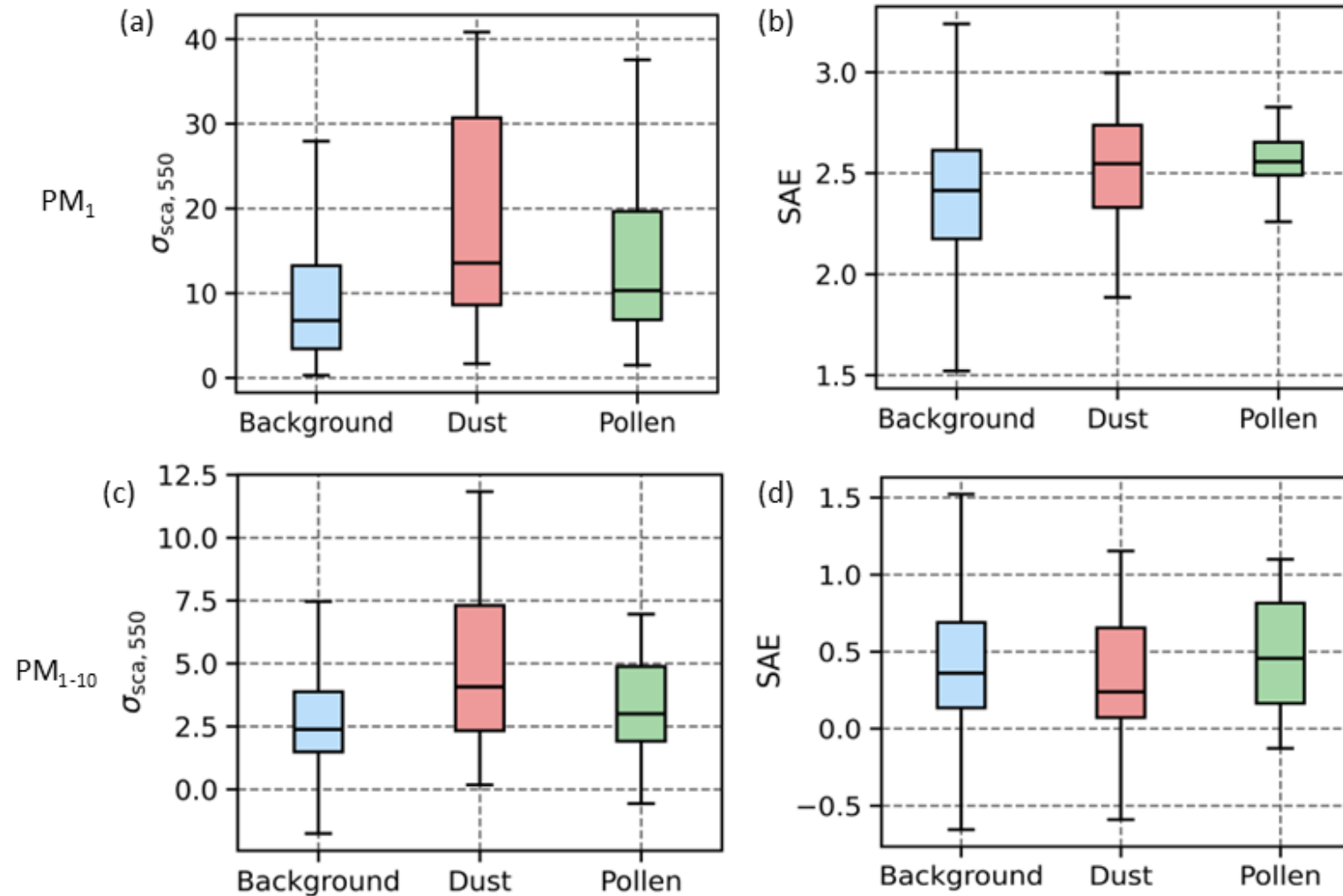
2

Which variables derived from the in-situ data of aerosol particles show a statistically significant difference between the non-event conditions, and dust/pollen events? (i.e.  $\sigma_{sca,550}$ , SAE, PM Mass, APSD metrics, etc.)

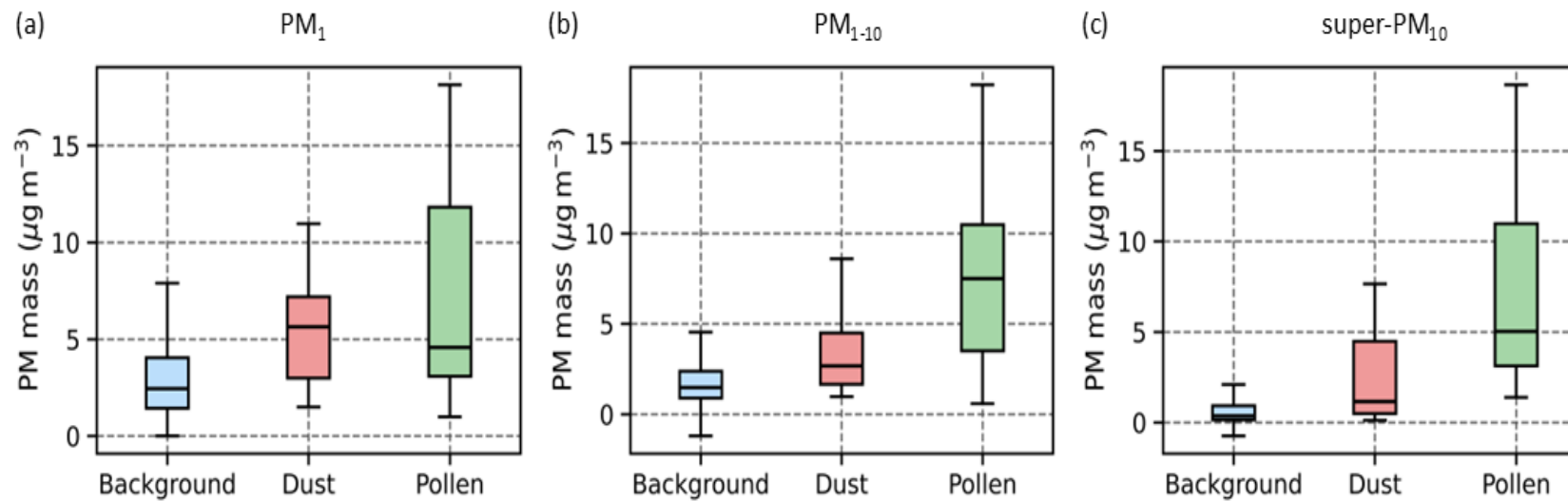
3

Which of these variables show such a difference, between the dust, and the pollen events?

# Effect on aerosol optical properties



# Effect on PM mass



# Research objectives

1

Effect of dust, and pollen events on optical properties, size distributions, and mass concentrations of aerosol particles.

2

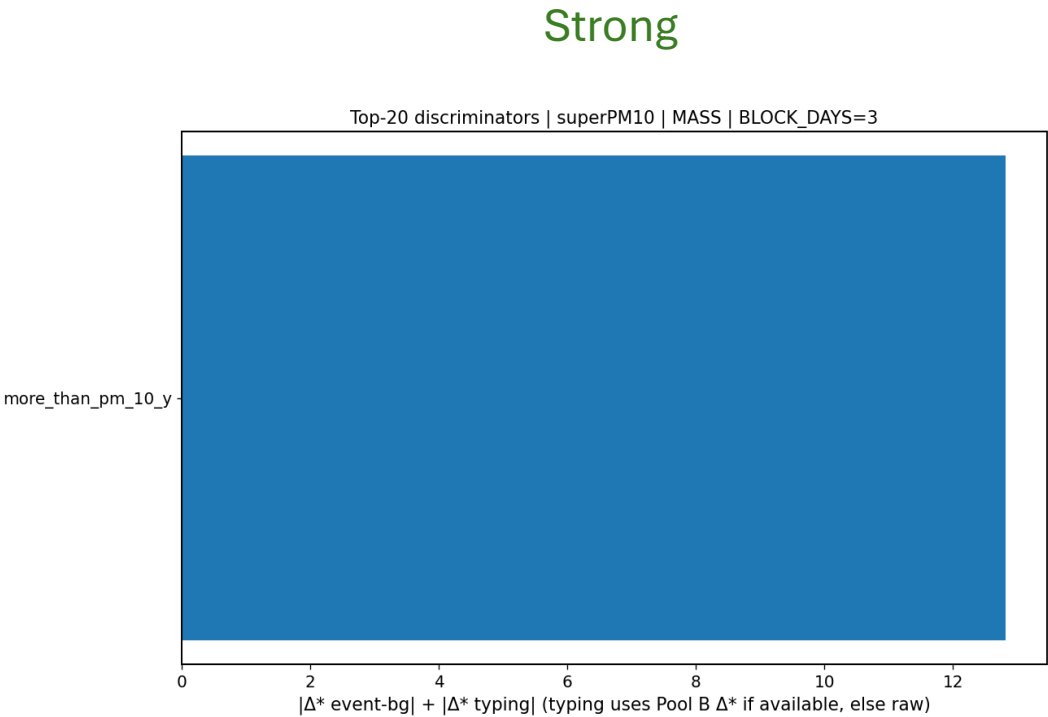
Which variables derived from the in-situ data of aerosol particles show a statistically significant difference between the non-event conditions, and dust/pollen events? (i.e.  $\sigma_{sca,550}$ , SAE, PM Mass, APSD metrics, etc.)

3

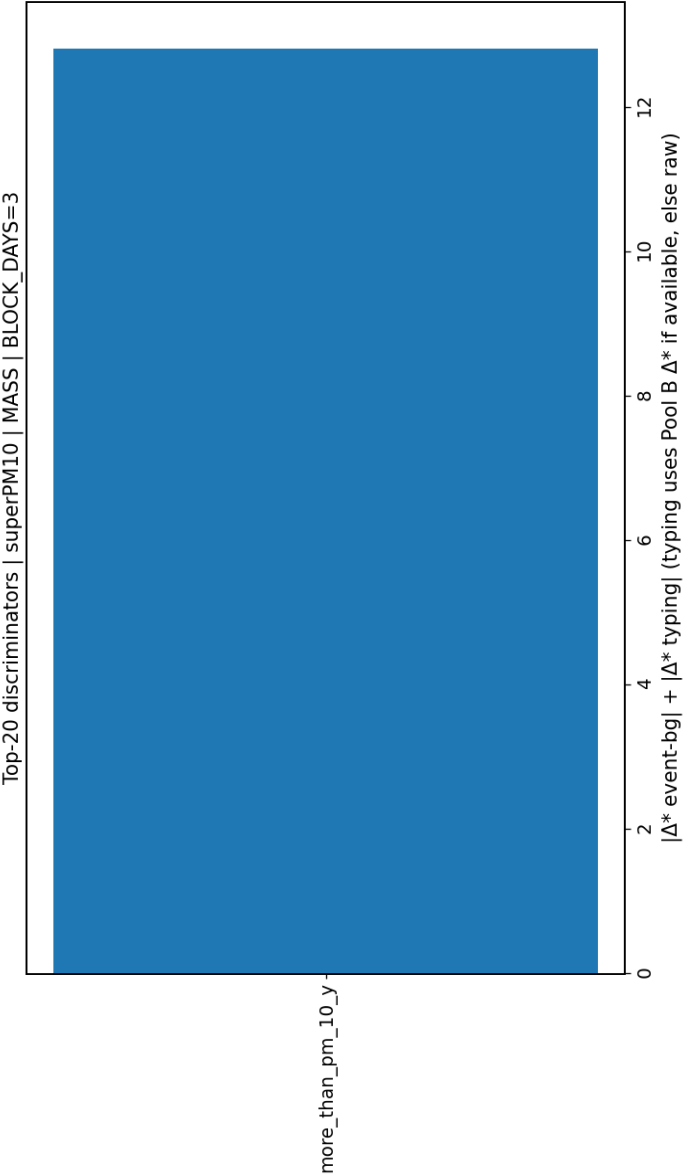
Which of these variables show such a difference, between the dust, and the pollen events?

# Ranking variables by separation strength

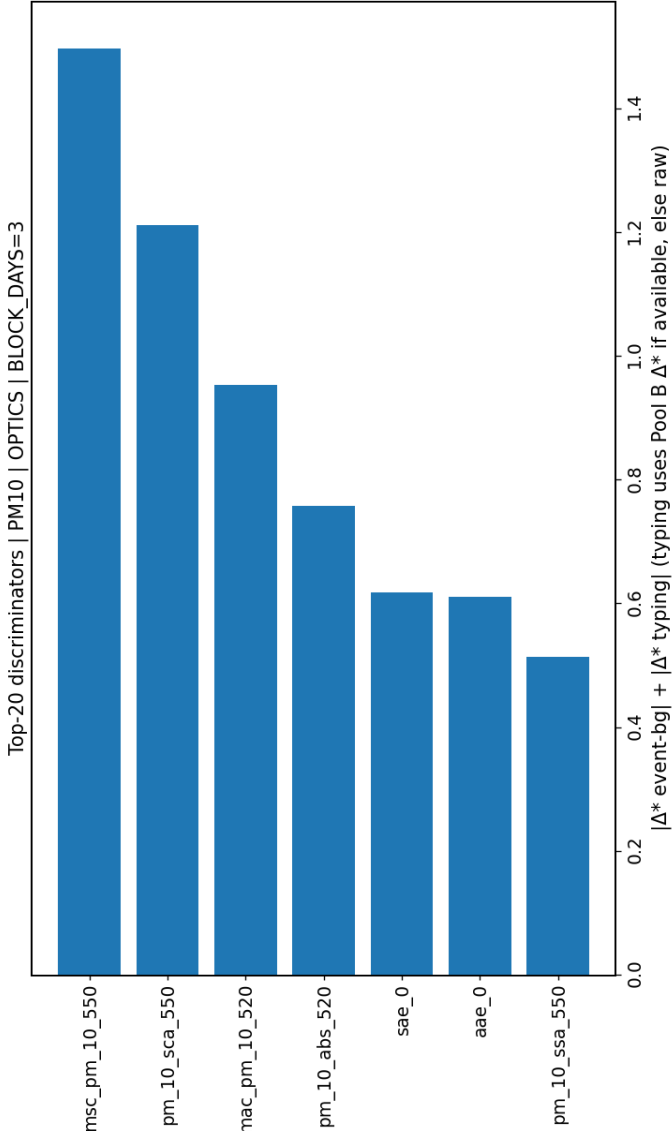
| Element                    | Meaning                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Title                      | Ranked subset of $PM_i$ variables $X_i$ ( $PM_i = PM_1, PM_{10}, PM_{1-10}, \text{super-}PM_{10}$ ; $X_i = \text{optical and/or mass variables}$ )                                                                                                                                                                                                                  |
| Y-axis                     | Individual variables $X_i$ (e.g., $\sigma_{\text{sca},550}$ , SAE, $PM_i$ mass)                                                                                                                                                                                                                                                                                     |
| X-axis                     | $S(X_i) =  \Delta_{\text{det}}^*(X_i)  +  \Delta_{\text{type}}^*(X_i) , \quad \text{where}$ $\Delta_{\text{det}}^*(X_i) = \frac{\tilde{X}_{i,\text{Event}} - \tilde{X}_{i,\text{Background}}}{\text{IQR}_{\text{Background}}}, \quad \Delta_{\text{type}}^*(X_i) = \frac{\tilde{X}_{i,\text{Dust}} - \tilde{X}_{i,\text{Pollen}}}{\text{IQR}_{\text{Background}}}.$ |
| Interpretation of $S(X_i)$ | <p><b>Weak:</b> <math>S(X_i) \in [0, 0.8)</math> (bottom 30%)</p> <p><b>Medium:</b> <math>S(X_i) \in [0.8, 2]</math> (middle 40%)</p> <p><b>Strong:</b> <math>S(X_i) \in (2, \infty)</math> (top 30%)</p>                                                                                                                                                           |



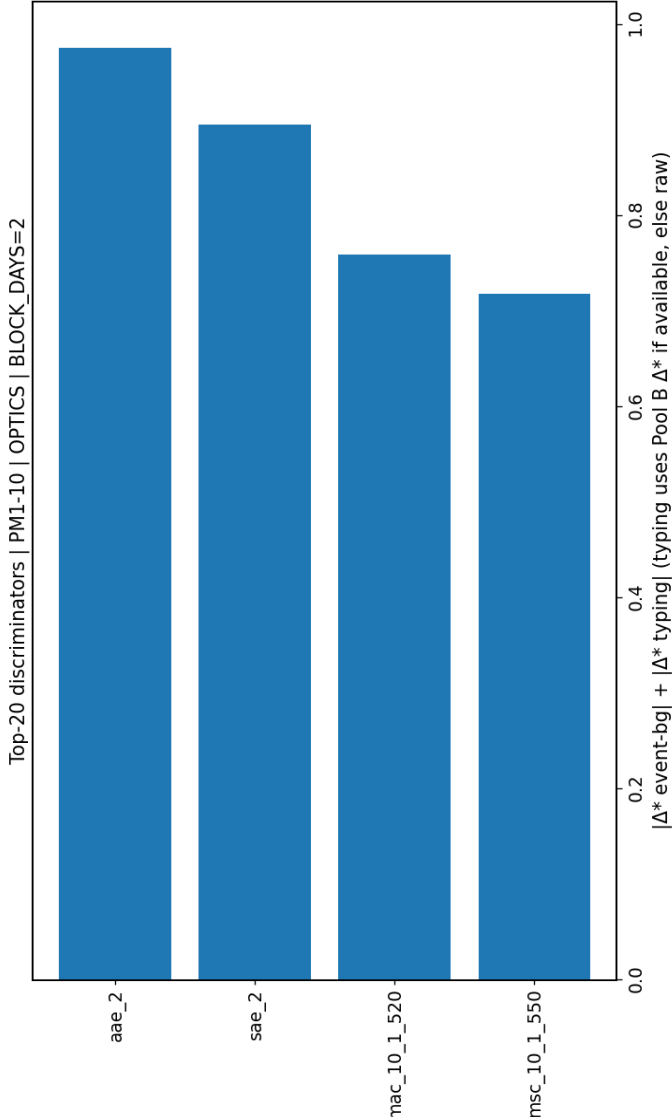
Strong



Medium

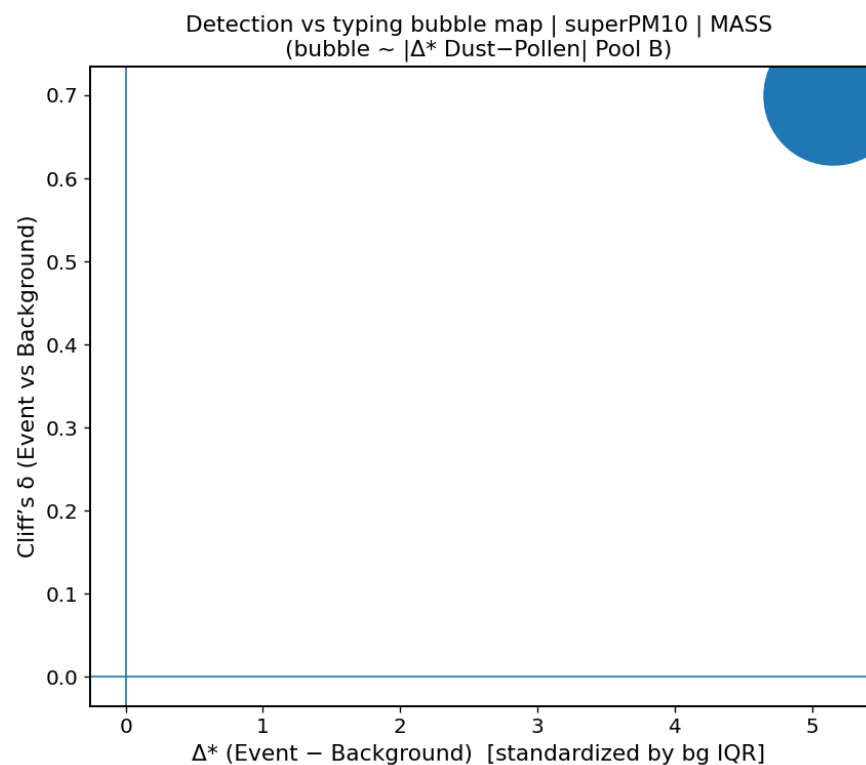


Weak



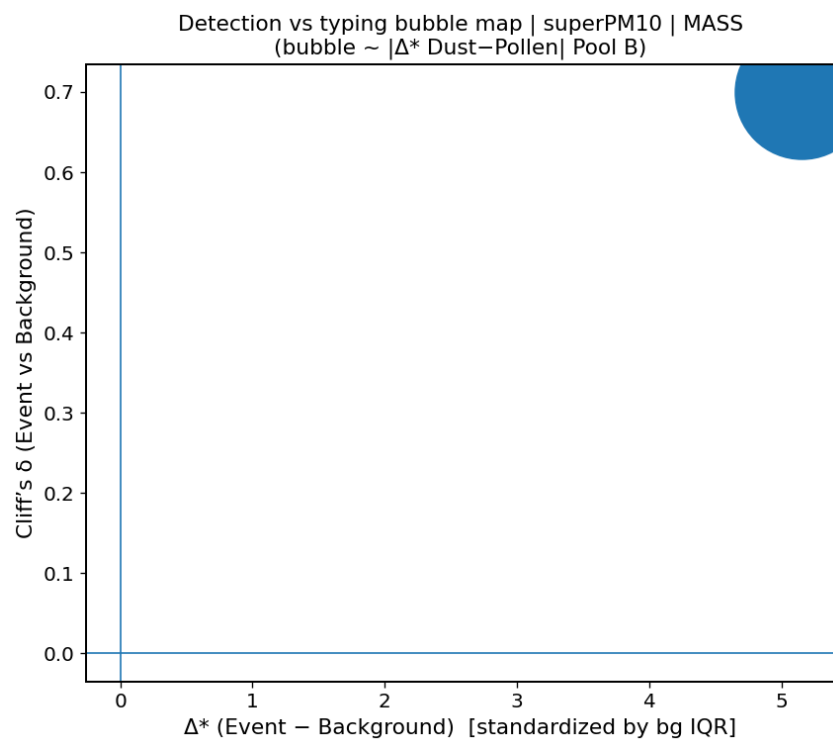
# Joint screening of event detection and event typing

Strong

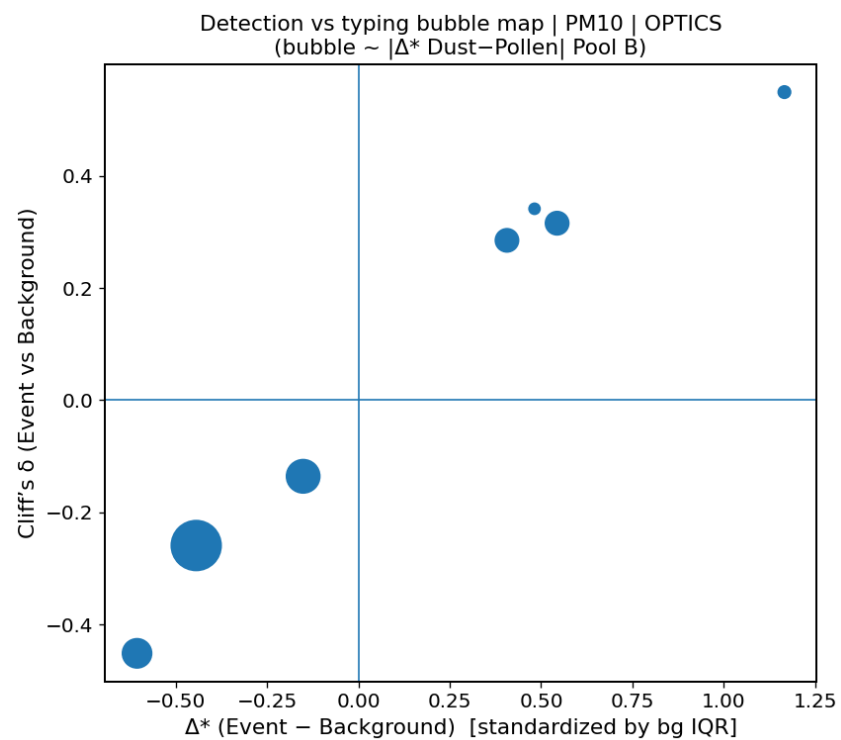


|                                                |                                                                                                                                                                                   |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>X-axis:</b> $\Delta^*_{\text{det}}$         | Strength of event detection (Event vs Background). Larger values indicate stronger deviation from background.                                                                     |
| <b>Y-axis:</b> Cliff's $\delta$                | Degree of separation between Event and Background distributions. Values closer to 1 indicate strong separation; values near 0 indicate weak separation.                           |
| <b>Bubble size:</b> $ \Delta^*_{\text{type}} $ | Strength of dust-pollen discrimination.<br><b>Large bubbles</b> indicate strong ability to distinguish Dust from Pollen.<br><b>Small bubbles</b> indicate weak typing capability. |
| <b>Top-right region</b>                        | Variables that strongly detect events and clearly separate them from background (high detection and strong separation).                                                           |
| <b>Near origin</b>                             | Variables with weak detection and poor separation; not useful discriminators.                                                                                                     |

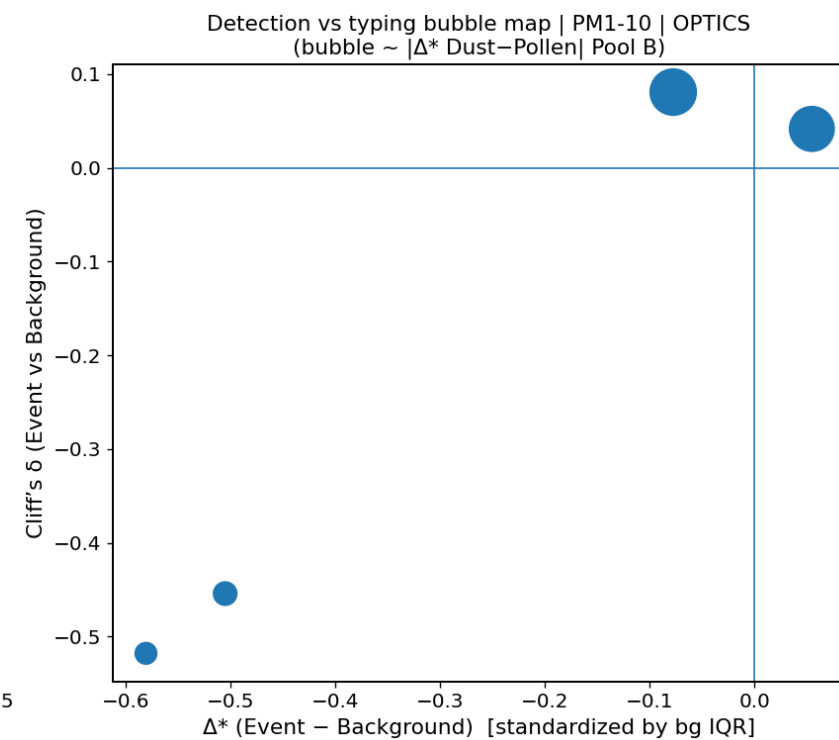
Strong



Medium



Weak



# Research objectives

1

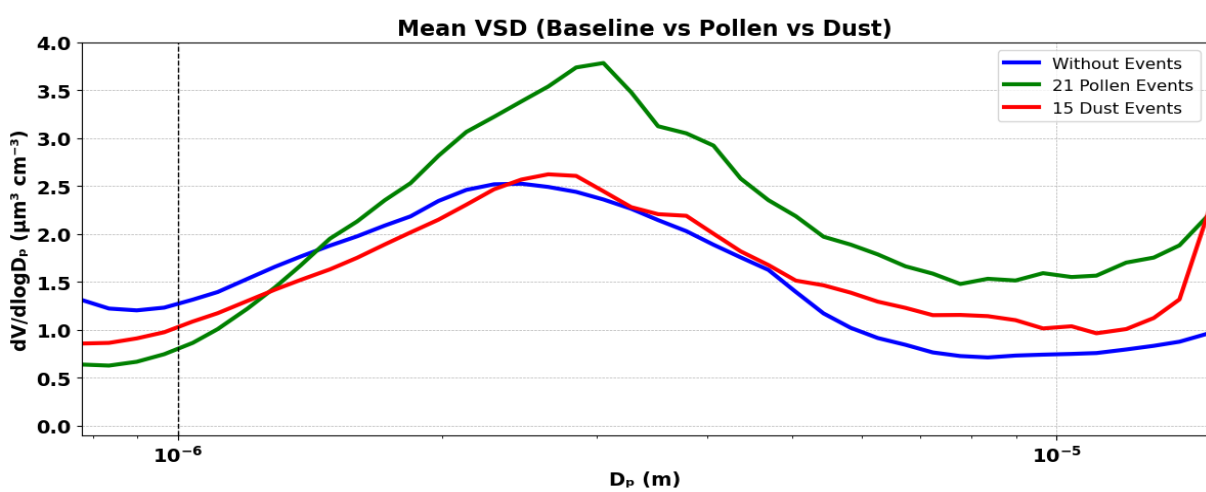
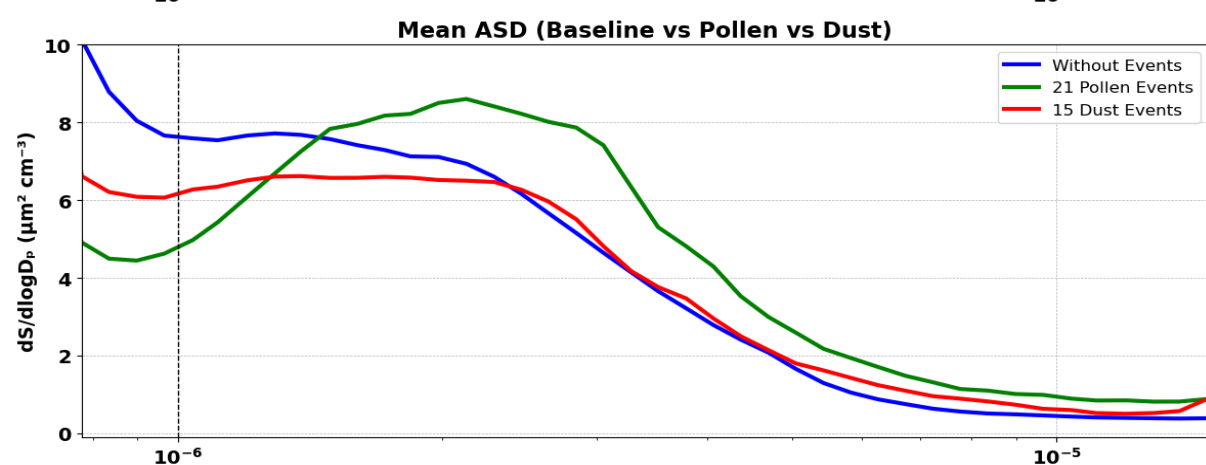
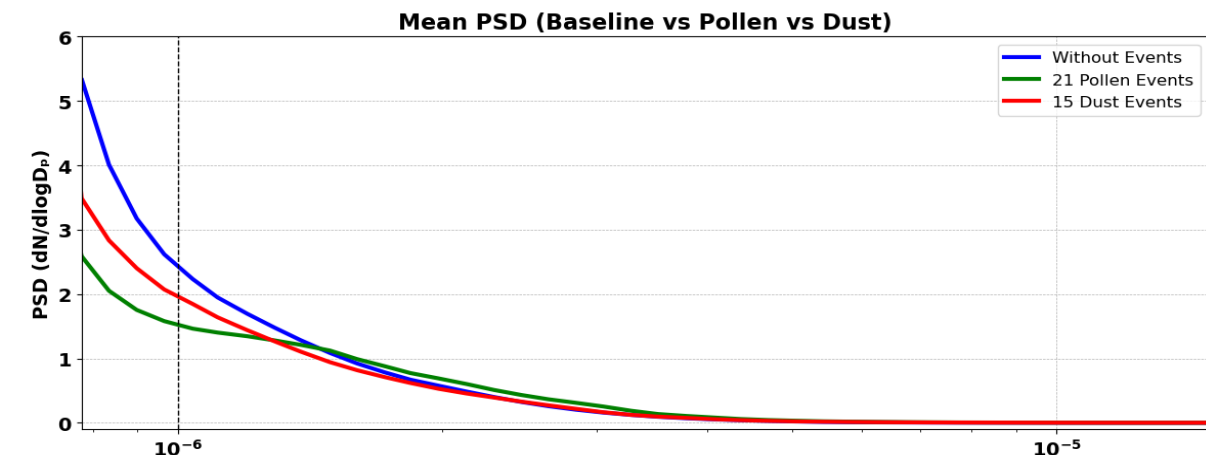
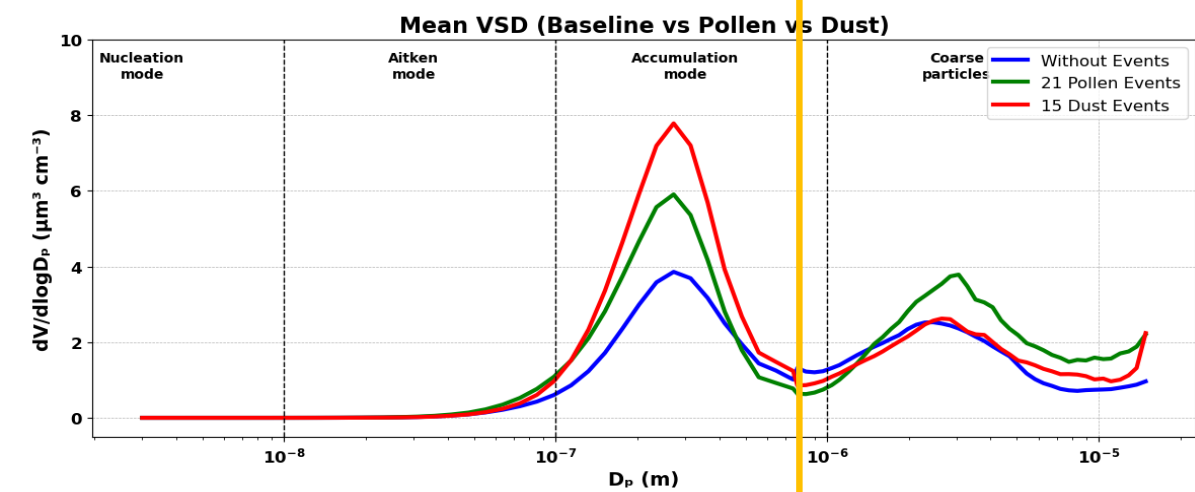
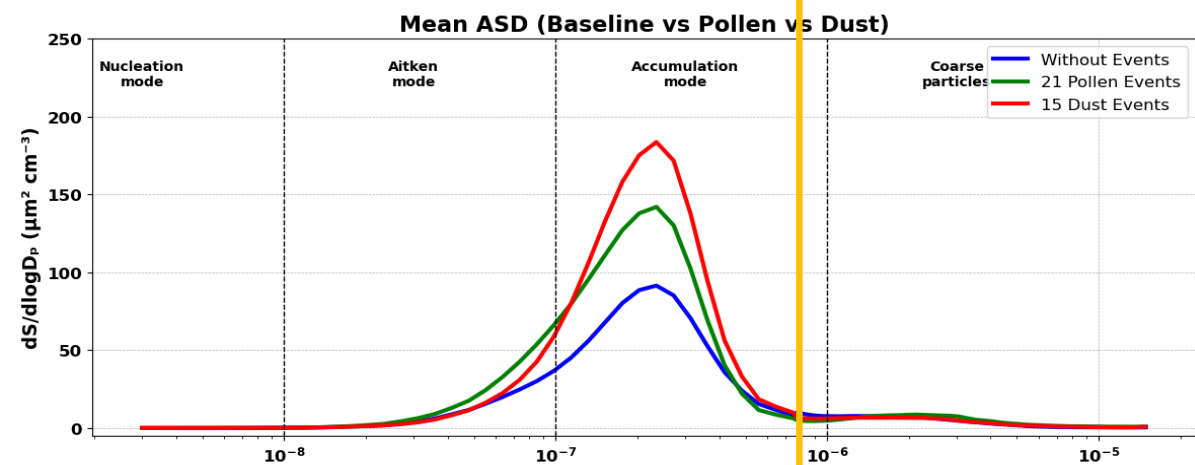
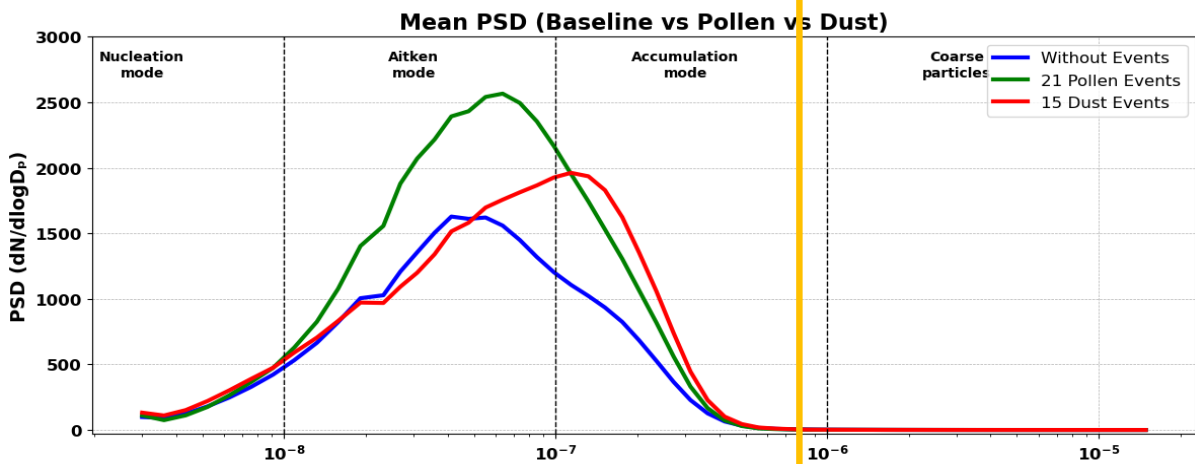
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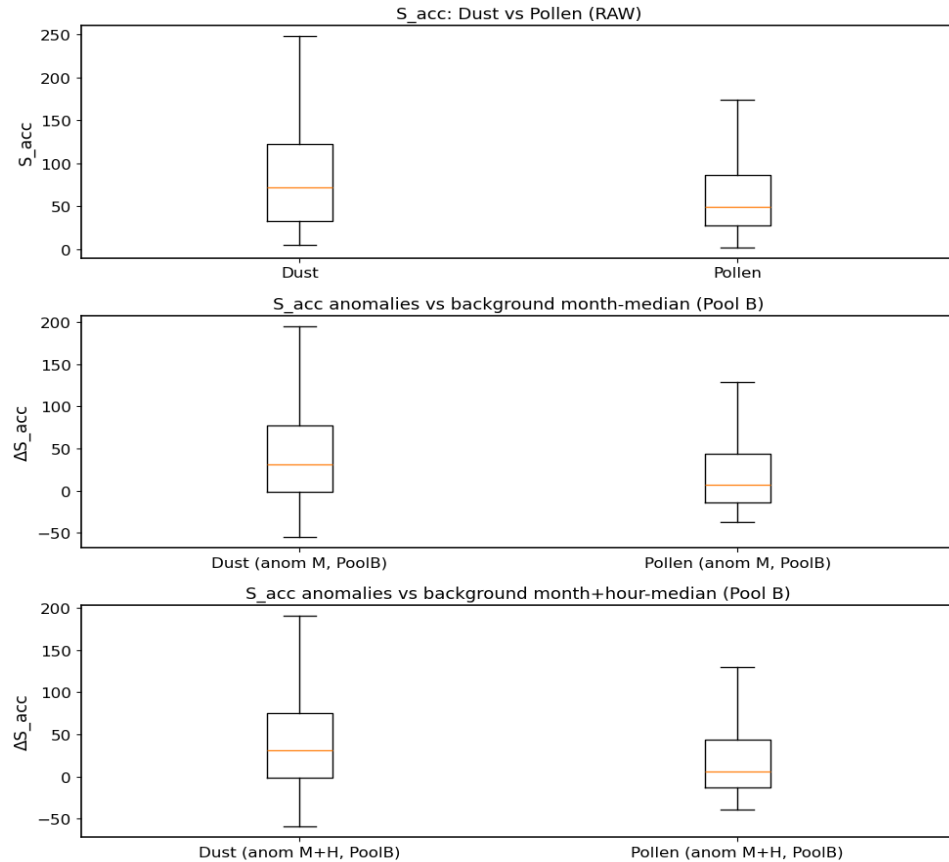
3

Which of these variables show such a difference, between the dust, and the pollen events?



# Confounding-controlled Dust–Pollen comparison

Strong



Row 1: Raw comparison

Direct comparison of  $X_{\text{dust}}$  and  $X_{\text{pollen}}$ . Represents apparent separation, which may reflect both intrinsic differences and temporal structure.

Row 2: Month-controlled anomalies

Seasonal effects removed using:

$$X'_M(t) = X(t) - \tilde{X}_{\text{background}|\text{month}(t)}$$

This removes the typical monthly background level and reduces seasonal confounding.

Row 3: Month + hour controlled anomalies

Seasonal and diurnal effects removed using:

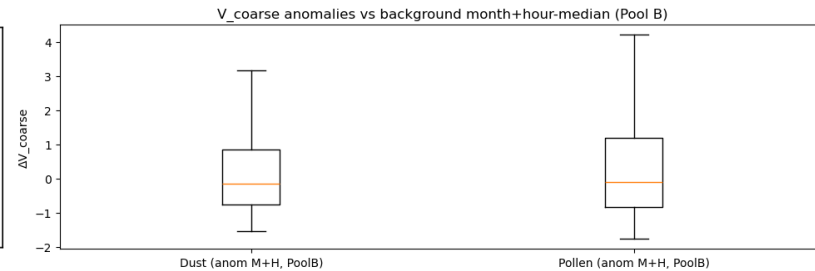
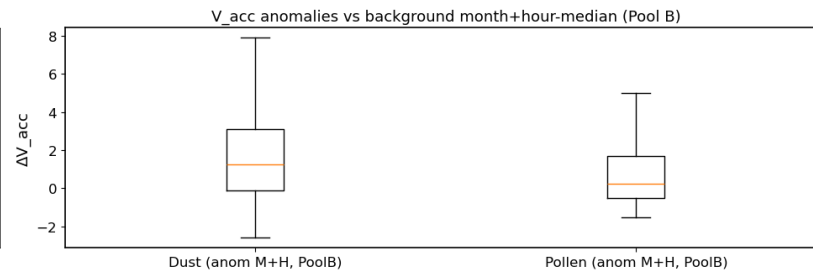
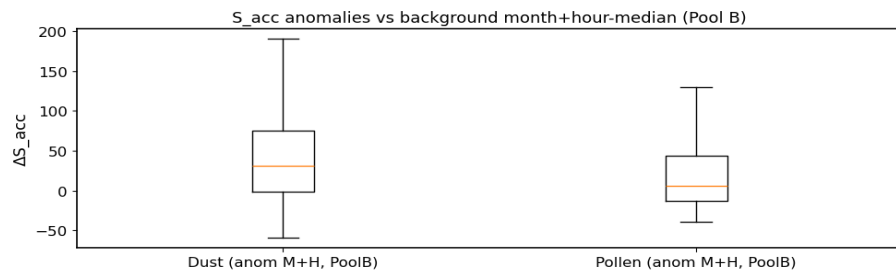
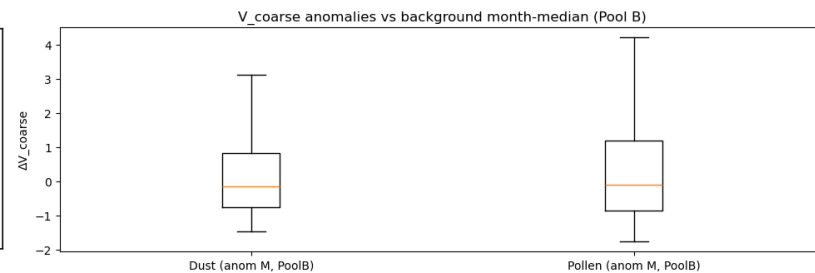
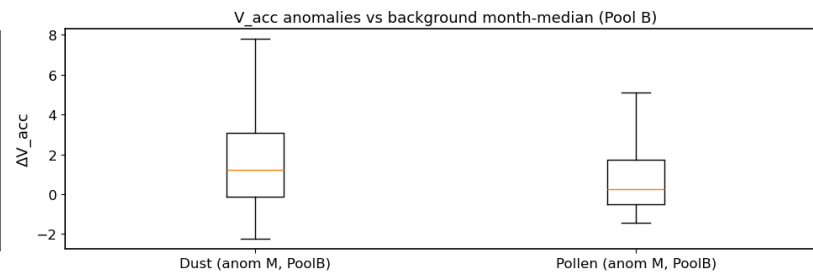
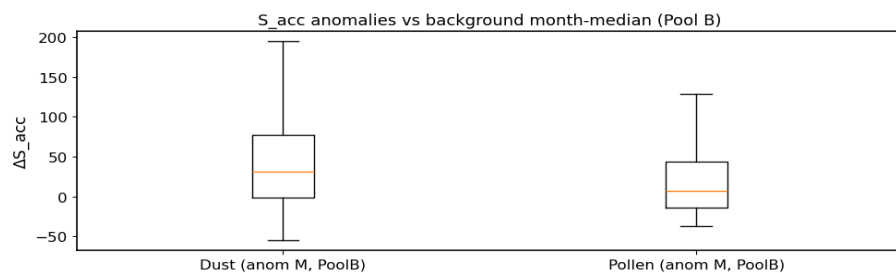
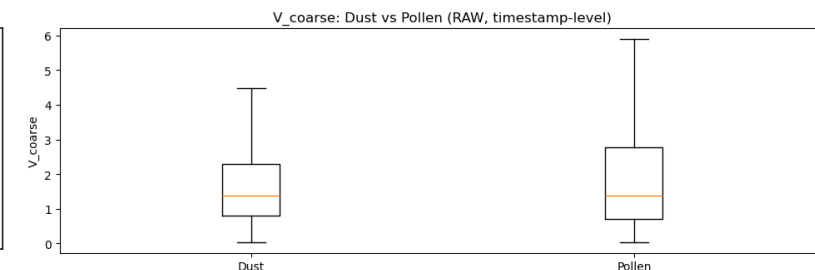
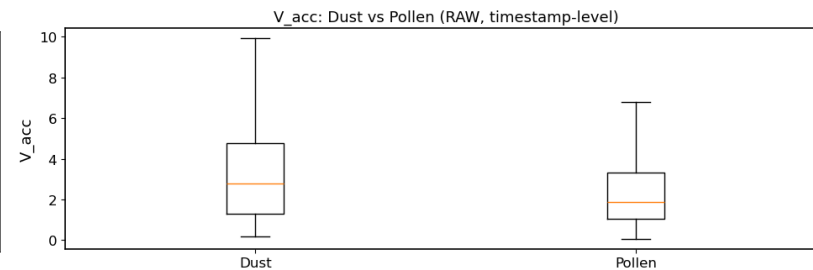
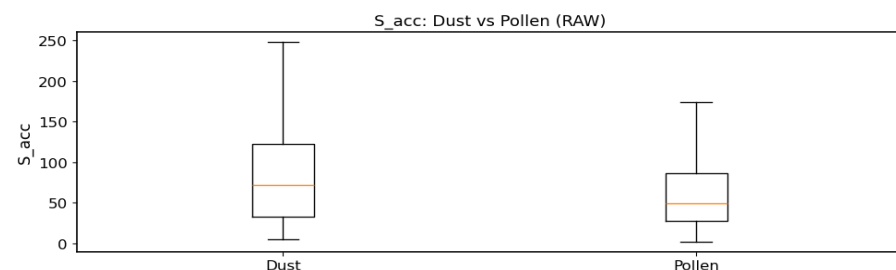
$$X'_{M+H}(t) = X(t) - \tilde{X}_{\text{background}|\text{month}(t), \text{hour}(t)}$$

This provides the strongest temporal control and isolates the most robust residual Dust–Pollen contrast.

Strong

Medium

Weak



Introduction

Methodology

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Conclusion

# Conclusions

1. Dust and pollen events are clearly detectable at SMEAR II because both enhance aerosol loading relative to non-event periods.
2. Bulk optical properties detect events well, but separate dust and pollen only weakly, whereas the clearest event-specific contrast appears in size-resolved PM mass, especially in the larger particle fractions.
3. Among PSD-derived variables, accumulation-mode surface area provides the clearest dust–pollen separation, showing that these event types are best distinguished with a size-resolved framework rather than with bulk optical properties alone.

# References

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3. Banerji, S., Luoma, K., Ylivinkka, I., Ahonen, L., Kerminen, V.-M., and Petäjä, T.: Measurement Report: Optical properties of supermicron aerosol particles in a boreal environment, *EGUsphere*, <https://doi.org/10.5194/egusphere-2025-1776>, 2025.
4. Eerdekens, G., Yassaa, N., Sinha, V., Aalto, P. P., Aufmhoff, H., Arnold, F., Fiedler, V., Kulmala, M., and Williams, J.: VOC measurements within a boreal forest during spring 2005: on the occurrence of elevated monoterpene concentrations during night time intense particle concentration events, *Atmos. Chem. Phys.*, 9, 8331–8350, <https://doi.org/10.5194/acp-9-8331-2009>, 2009.



Thank you for listening!